

Role C · Map, Ranking & Cockpit UI: Agent Orchestration Plan

UrbanTwin: Market-Sim · Claude Impact Lab #2 · Build window 13:00–20:00 Owner: Mahesh (orchestrator) · Owns `src/ui/` and the pre-event scaffold Source of truth for scope, contract and gates: `IMPLEMENTATION_PLAN.md` (\$1 tiers, \$5 contract, \$7 role C, \$10 gates)

Done-check for role C (from \$7): a weight change reorders the ranking in under 100 ms, and the map holds 60 fps with 100 trails.

0. Ground truth checked on this machine (2026-10-05)

Item	Finding	Consequence for the plan
Repo	<code>claude_impact_lab</code> is its own Git repo on main (dcac2bb), remote <code>gi60wedo/claude_impact_lab</code> . It has no <code>package.json</code> , no <code>src/</code> , no <code>public/</code> .	The whole scaffold (C0) is still to do.
<code>.scratch/</code>	Not listed in <code>.git/info/exclude</code> .	Add it before the first <code>ruddr</code> run so briefs and run state stay out of diffs.
<code>datasets/dop20/</code>	4 JPEG quadrants, each 4200 × 4200 px, 1050 m × 1050 m (0.25 m/px). The filename holds the bbox <code>minx_miny_maxx_maxy</code> in EPSG:25832. Total 12 MB.	Downscale to 2100 px (0.5 m/px) for GPU memory. Convert the 4 corners of each quadrant to WGS84 and use the 4-corner bounds form of <code>BitmapLayer</code> . A UTM rectangle is slightly rotated in lng/lat ($\approx 1.5^\circ$ convergence at 11.08° E, zone 32 meridian is 9° E), so a 2-corner bbox would misalign by several metres at the edges.
<code>datasets/alkis/</code>	4 transparent PNG overlays with the same bbox naming.	Optional overlay with the same corner transform.
<code>datasets/lod2/</code>	4 CityGML tiles, 525 MB total, <code>bldg:measuredHeight</code> present, ~6,760 buildings in one tile, CRS ETRS89/UTM32.	The browser cannot load this. Somebody must convert it to a small <code>buildings.json</code> (footprint + height).
<code>datasets/osm/altstadt.json</code>	Overpass JSON (<code>elements[]</code> with inline geometry): 3,908 building ways (2,013 with <code>building:levels</code> , 182 with height), 224 bollards, 4,148 highway features.	Fallback for 3D buildings: OSM footprints, height = height or <code>building:levels</code> × 3 m or 9 m. Bollard markers come straight from OSM nodes.
Node / tools	node v26.4.0, npx, bun; no npnm. Playwright browsers cached in <code>~/.cache/ms-playwright</code> (chromium-1243), but the playwright package is not installed in this repo.	Use npm. Add <code>@playwright/test</code> as a dev dependency in C0.
codex	<code>codex-cli 0.160.0</code> . <code>~/.codex/config.toml</code> : <code>model = "gpt-6.1-sol"</code> , <code>model_reasoning_effort = "medium"</code> , <code>sandbox_mode = "danger-full-access"</code> , <code>approval_policy = "never"</code> .	Codex defaults to full access with no approvals. Always pass a sandbox explicitly.
claude	2.1.289 (Claude Code).	Headless mode works with <code>-p</code> , <code>--model</code> , <code>--effort</code> , <code>--permission-mode</code> , <code>--max-budget-usd</code> , <code>-w/--worktree</code> .
ruddr	<code>ruddr 0.6.3</code> , skill at <code>~/.claude/skills/ruddr-delegate/SKILL.md</code> . Model catalog (<code>ruddr models --json</code>): Codex <code>gpt-6.1-sol</code> (efforts low...ultra), <code>gpt-6-luna</code> , others; Claude <code>claude-opus-5-5</code> (default), <code>claude-fable-5-1</code> , <code>claude-haiku-4-5-20251001</code> , <code>claude-sonnet-5</code> . claude-sonnet-5-5 is not in the Ruddr catalog , and the catalog lists no efforts for Claude models.	Run <code>ruddr models add claude claude-sonnet-5-5 --label "Claude Sonnet 5.5"</code> once before the event, then confirm with <code>ruddr models --json</code> . If a Claude <code>--effort</code> is rejected by Ruddr, drop the flag or launch that agent with <code>claude -p</code> instead.

1. Design choices that keep the task tree small

1. **One store, three adapters, one layout file. Mahesh writes these by hand.** They are about 120 lines total and they are the only files that every subtask touches. Agents may import them and may not edit them.
2. `src/ui/state/store.ts`: a tiny `useSyncExternalStore` store (no new dependency). It holds `candidates`, `results`: `Record<candidateId, SimulationResult>`, `weights`, `scenario`, `slice`, `selectedId`, `brief`, `playing`, `timeSec`.
3. `src/ui/adapters/types.ts`: `SimClient { run(candidateId, scenario, mitigations, seed): Promise<SimulationResult> }` and `BriefClient { brief(input): Promise<Brief> }`.
4. `src/ui/adapters/fixtures.ts`: both clients backed by `public/data/mock-*.json`.
5. `src/ui/App.tsx`: a CSS grid with four named slots (`map`, `ranking`, `whatif`, `cockpit`).
6. **The UI never computes a criterion.** It reads `SimulationResult.criteria` and computes only the weighted sum with the active weights. Until D's `src/score/` exports a function, C uses a 10-line `weightedScore()` in `src/ui/ranking/applyWeights.ts` and swaps the import at 17:00.
7. **Base map without vector tiles.** The DOP20 bitmap is the base. `MapLibre` runs with an empty style (`{version: 8, sources: {}, layers: [{id:'bg', type:'background', paint:{'background-color':'#111'}}]}`) and `deck.gl` draws on top through `@deck.gl/mapbox MapboxOverlay` in interleaved mode. The demo then works offline.
8. **Static C assets live in `public/base/`, owned by C.** A owns `public/data/`. C never writes there except for the fixtures committed jointly at 13:20.
9. **One `deck.gl` version.** All `@deck.gl/*` packages are pinned to the same exact version in `package.json`, with no caret.

2. Task tree

Tier tags follow §1 of the master plan: **M** = Must, **S** = Should, **X** = Stretch (skipped unless all M and S items are merged by 16:30).

2.1 Pre-event (before 13:00)

ID	Tier	Subtask	Files	Depends on	Done-condition (orchestrator runs it)
C0	M	Repo scaffold: Vite + React + TS, Tailwind, deck.gl + MapLibre, Vitest, Playwright, ESLint off. <code>src/contracts.ts</code> with the §5 types pasted as a <i>draft</i> (freeze at 13:20). Empty folders <code>src/rank/</code> <code>src/sim/</code> <code>src/score/</code> <code>server/</code> with <code>.gitkeep</code> . <code>npm run dev, build, test, e2e</code> scripts.	<code>package.json</code> , <code>package-lock.json</code> , <code>vite.config.ts</code> , <code>tsconfig*.json</code> , <code>index.html</code> , <code>src/main.tsx</code> , <code>src/index.css</code> , <code>src/contracts.ts</code> , <code>playwright.config.ts</code> , <code>.gitignore</code>	none	<code>npm ci && npm run build && npm test -- --run && npx tsc --noEmit -p tsconfig.app.json all exit 0; npm ls @deck.gl/core prints exactly one version; git grep -nE 'sk-ant ANTHROPIC_API_KEY=' -- . ':!*.md' prints nothing.</code>
C1	M	Base assets: script converts the 4 DOP20 quadrants to 2100 px JPEGs plus <code>public/base/dop20.json</code> (<code>[{url, bounds:[[lng,lat]x4}]</code>), corners in order bottom-left, top-left, top-right, bottom-right). Same for ALKIS PNGs. Fallback <code>public/base/buildings-osm.json</code> from OSM building ways.	<code>tools/base_assets.py</code> , <code>public/base/*</code>	C0	<code>uv run --with pyproj --with pillow python tools/base_assets.py</code> exits 0; <code>public/base/</code> < 15 MB; a Python check confirms Lorenzkirche 11.07773, 49.45095 falls inside exactly one quadrant's corner polygon; building count ≥ 3,000.

ID	Tier	Subtask	Files	Depends on	Done-condition (orchestrator runs it)
C2	M	UI fixtures that match \$5 exactly: 18 candidates (3 benchmarks, ≥ 5 rejected with <code>rejectReason</code>), 3 <code>SimulationResults</code> $\times$ <code>SUNNY_SAT/RAINY_SAT</code> , ≥ 100 trips along real OSM footways, 1 Brief. A Vitest file type-checks every fixture against <code>contracts.ts</code> .	<code>src/ui/__fixtures__/_gen.ts</code> , <code>src/ui/__fixtures__/*.json</code> , <code>src/ui/__fixtures__/_fixtures.test.ts</code>	C0, C1	<code>npm test -- --run src/ui/__fixtures__</code> passes; <code>npx tsc --noEmit</code> passes with <code>resolveJsonModule</code> ; trip count ≥ 100 per result. These become <code>public/data/mock-*.json</code> at 13:20 if B and D have nothing better.
C3a	M	Orchestrator files from \$1.1 (store, adapters, App layout).	<code>src/ui/state/store.ts</code> , <code>src/ui/adapters/*</code> , <code>src/ui/App.tsx</code>	C0	Maresh writes them. <code>npm run build</code> passes.

Pre-event agent ask to A: `public/data/buildings.json = [{polygon:[[lng,lat]...], h:number}]` from LoD2 `measuredHeight`. If it is not committed by 13:00, C uses `public/base/buildings-osm.json` and swaps later by changing one URL.

2.2 13:00-15:00 (gate 15:00: "UI shows fixtures end to end")

ID	Tier	Subtask	Files	Depends on	Done-condition
C4	M	Map shell and static layers: MapLibre + MapboxOverlay, DOP20 BitmapLayers, extruded buildings (<code>SolidPolygonLayer</code> or <code>PolygonLayer</code> with extruded), candidate polygons colored by rank, rejected ones grey with a tooltip that shows <code>rejectReason</code> , click selects a candidate. Bollards from OSM as <code>ScatterplotLayer</code> . Attribution line from \$3.	<code>src/ui/map/**</code>	C0-C3a	<code>npm run e2e -- map.spec.ts</code> passes: canvas visible, no console errors, tooltip with the reject reason appears on hover of a rejected candidate; screenshot <code>e2e/_shots_/map.png</code> saved and inspected by Maresh (aerial photo aligns with building extrusions at Lorenzkirche).
C5	M + S	Ranking panel: MarketScore bars sorted by weighted score, 5 weight sliders with live renormalization (sum = 1), 3 presets from \$6, guard badge "fails a stakeholder group" when any persona < 40, rejected list collapsed below.	<code>src/ui/ranking/**</code>	C3a	Vitest: <code>applyWeights</code> renormalizes and reorders on preset change, guard flag fires below 40. Playwright <code>ranking.spec.ts</code> : sets the accessibility slider, measures <code>performance.now()</code> from input to the DOM order change via <code>MutationObserver</code> , median of 20 changes < 100 ms.
C6	M	Cockpit: consensus dial (min and mean of persona scores), 4 persona cards (score, topFriction chip, verdict), brief panel rendering why, comparisons, losers, buttons Apply Claude mitigation (calls <code>SimClient.run</code> with <code>losers[*].mitigation</code>) and Export council brief (downloads <code>councilBriefMd</code> as <code>.md</code>).	<code>src/ui/cockpit/**</code>	C3a	Playwright <code>cockpit.spec.ts</code> : 4 cards render the fixture scores verbatim; clicking Export yields a download whose text equals <code>brief.councilBriefMd</code> ; clicking Apply calls the adapter once (spy) and re-renders the cards.

2.3 15:00-16:30 (dynamics)

ID	Tier	Subtask	Files	Depends on	Done-condition
C7	M + S	Simulation layers: TripsLayer with persona colours (senior purple, vendor orange, commuter blue, retailer grey), HeatmapLayer from bySlice[slice].heat, bottleneck markers sized by severity, kiosk markers when mitigations is non-empty. Time-slice slider (05:30 / 11:30 / 15:00) and play/pause driving timeSec through one requestAnimationFrame loop. In-app FPS probe on window.__fps when the URL has ?perf=1.	src/ui/map/simLayers.ts, src/ui/controls/**, src/ui/perf/fps.ts	C4	perf.spec.ts run headed on the demo laptop: ?perf=1 with 100 trails playing, median of window.__fps over 10 s ≥ 55 , 3 runs out of 3. Screenshot e2e/___shots___/trails.png shows trails and heatmap.
C8	M + S	What-if bar: Sunny / Rainy / Christmas market buttons that call SimClient.run for the shortlist and show a spinner; side-by-side compare of two selected candidates (S).	src/ui/whatif/**, src/ui/compare/**	C3a, C5	Playwright whatif.spec.ts: clicking Rainy loads RAINY_SAT fixtures and changes at least one bar value; compare view shows both candidates' 5 criteria.
C9	X	Draw a custom site (EditableGeoJsonLayer is out; use 4 clicks → polygon).	src/ui/draw/**	all M	Only started if C4-C8 are merged by 16:30.

2.4 17:00-18:00 (integration, freeze 18:00)

ID	Tier	Subtask	Files	Depends on	Done-condition
C10	M	Swap adapters: adapters/worker.ts wraps B's Web Worker; adapters/http.ts POSTs to /api/brief; import D's MarketScore in place of applyWeights. Feature flag ?data=fixtures keeps the fixture path alive.	src/ui/adapters/**, one import line in src/ui/ranking/applyWeights.ts	C4-C8, B worker, D route	Full demo path (§11 pitch order) runs 3 times in a row in demo.spec.ts with real adapters; ?data=fixtures still passes all specs.
C11	M	Projector polish: contrast, font size ≥ 16 px, persona colours readable on the aerial photo.	src/index.css, src/ui/** styles	C10	Mahesh checks on the projector at 18:00.

3. Agent assignment

Provider facts used below come from ruddr models --json and the CLI help on this machine.

ID	Agent	Model	Effort	Why
C0	Codex	gpt-6.1-sol	medium	Well-specified tooling work with a crisp command-based done-check. Codex handles package and config setup reliably.
C1	Claude	claude-sonnet-5-5	medium	Short Python script with a geodesy detail (corner order, rotation). Sonnet handles it cheaply; the done-check catches a wrong transform.
C2	Claude	claude-haiku-4-5-20251001	(none)	Mechanical fixture generation against a fixed type. Cheapest model is enough; tsc catches drift.
C3a	Mahesh	—	—	Shared files. Hand-written so no agent owns them.
C4	Claude	claude-opus-5-5	high	deck.gl layer design, MapLibre interleaving and coordinate alignment carry the most architecture risk in role C.
C5	Codex	gpt-6.1-sol	high	Self-contained component with a hard perf target and a measurable test.
C6	Claude	claude-sonnet-5-5	medium	Standard React components rendering given data.

ID	Agent	Model	Effort	Why
C7	Codex	gpt-6.1-sol	high	Performance-critical animation loop and layer tuning against an fps number.
C8	Claude	claude-sonnet-5-5	medium	Small UI with adapter calls.
C9	Claude	claude-sonnet-5-5	medium	Only if time remains.
C10	Mahesh + Claude interactive	claude-opus-5-5	high	Integration debugging across A/B/D code needs a human in the loop. Run interactively, not as a swarm job.
C11	Mahesh	—	—	Visual judgement on real hardware.
Review of Codex branches	Claude	claude-sonnet-5-5	medium	Other-vendor review, read-only (--permission-mode plan).
Review of Claude branches	Codex	gpt-6.1-sol (config default)	medium	codex review --base c/ui.

Table 5.1 (claude-fable-5-1) is not assigned. It stays as a spare vendor-internal fallback when Opus stalls on C4 or C10.

4. Parallel schedule, worktrees and ownership

Worktrees sit next to the repo so npm installs do not collide. Each one needs its own npm ci (about 1 minute; run it in the background).

Window	Parallel agents	Worktree (../ut-<id>) and branch	Owns (may write)
Pre-event, step 1	C0 alone	../ut-c0 · c/c0-scaffold	root config files, src/contracts.ts, src/main.tsx, src/index.css
Pre-event, step 2	C1 C2 (C2 starts with OSM, reads C1 output when ready)	../ut-c1 · c/c1-base; ../ut-c2 · c/c2-fixtures	C1: tools/, public/base/. C2: src/ui/__fixtures__/
13:30–15:00	C4 C5 C6	../ut-c4 · c/c4-map; ../ut-c5 · c/c5-ranking; ../ut-c6 · c/c6-cockpit	C4: src/ui/map/, e2e/map.spec.ts. C5: src/ui/ranking/, e2e/ranking.spec.ts. C6: src/ui/cockpit/, e2e/cockpit.spec.ts
15:00–16:30	C7 C8	../ut-c7 · c/c7-sim; ../ut-c8 · c/c8-whatif	C7: src/ui/map/simLayers.ts, src/ui/controls/, src/ui/perf/, e2e/perf.spec.ts. C8: src/ui/whatif/, src/ui/compare/, e2e/whatif.spec.ts
17:00–18:00	C10 (interactive)	main checkout · c/c10-integrate	src/ui/adapters/

Read-only for every agent: src/contracts.ts, src/ui/state/store.ts, src/ui/adapters/types.ts, src/ui/App.tsx, package.json, package-lock.json, tailwind/theme tokens in src/index.css, everything outside src/ui/. If an agent needs a new dependency or store field, it writes // TODO(subagent): need X and stops that part. Mahesh adds it on main and tells the agent to rebase with ruddr steer.

Mounting: each component exports one default component (<MapView/>, <RankingPanel/>, <Cockpit/>, <WhatIfBar/>). App.tsx already imports all four from day one with stub files committed in C3a, so merges never touch App.tsx.

Merge order and who merges: Mahesh merges C branches into his integration branch c/ui in the order C0 → C1 → C2 → C3a → C5 → C6 → C4 → C8 → C7 (cheapest-to-verify first, map last because it is the slowest

to review). After each merge he runs `npm ci && npm run build && npm test -- --run && npm run e2e`. D merges c/ui into main (§12 of the master plan) at 15:00 and at 17:00.

5. Prompts

5.1 Shared preamble (prepend to every brief)

Project: UrbanTwin Market-Sim, a Vite + React + TypeScript app with deck.gl over MapLibre. You work in a git worktree on branch {BRANCH}. Other agents edit other folders of the same project at the same time. Read IMPLEMENTATION_PLAN.md §5 (data contract) and §7 role C before editing.

Contract rules:

- Import types only from src/contracts.ts. Do not add, rename or make optional any field. If you need data the contract lacks, write ``// TODO(subagent): contract needs <field>`` and work around it.
- Never hard-code a score, percentage or criterion value in UI code. Render values from the data.
- Read app state only through src/ui/state/store.ts and data only through src/ui/adapters/types.ts.

Scope fence: you may create or edit files ONLY under {ALLOWED}. Do not modify files outside {ALLOWED}. Do not edit package.json, package-lock.json, src/contracts.ts, src/ui/App.tsx, src/ui/state/*, src/ui/adapters/*, src/index.css. Do not add dependencies. Do not touch .scratch/ (controller files).

Styling: Tailwind utility classes only, using the existing tokens in src/index.css (dark theme, persona colours senior #8b5cf6, vendor #f97316, commuter #3b82f6, retailer #9ca3af). No new CSS files, no other UI library.

Secrets: never read, print or write .env or any API key. No network calls except to localhost.

Commit policy: leave all changes uncommitted. The orchestrator reviews and commits. Dependencies are already installed (npm ci ran). The sandbox may block network; that is expected.

5.2 Task blocks (append to the preamble, then append the Ruddr handoff block verbatim)

C0 · scaffold (Codex). {ALLOWED} = repo root config files, src/main.tsx, src/index.css, src/contracts.ts, src/ui/, e2e/, empty .gitkeep folders. This brief overrides the preamble's ban on package.json.

Goal: scaffold the app. Use npm (no pnpm). Vite + React + TS, Tailwind, Vitest (jsdom), @playwright/test, maplibre-gl, deck.gl packages @deck.gl/core @deck.gl/layers @deck.gl/geo-layers @deck.gl/aggregation-layers @deck.gl/mapbox @deck.gl/react, all pinned to ONE exact identical version (no ^ or ~). Paste the TypeScript block from IMPLEMENTATION_PLAN.md §5 into src/contracts.ts verbatim with a header comment "DRAFT until 13:20 freeze". Create src/rank src/sim src/score server with .gitkeep. Scripts: dev, build (tsc -b && vite build), test (vitest), e2e (playwright test). Playwright config: baseURL http://localhost:5173, webServer = npm run dev, chromium only. src/index.css: Tailwind import plus CSS variables for the dark theme and the four persona colours listed above. Done-check (run all): npm ci; npm run build; npm test -- --run; npm ls @deck.gl/core (exactly one version).

C1 · base assets (Claude Sonnet). {ALLOWED} = tools/, public/base/.

Goal: tools/base_assets.py produces browser assets from datasets/.

1. For each datasets/dop20/*.jpg and datasets/alkis/*.png: parse minx_miny_maxx_maxy (EPSG:25832) from the name,
 - resize to 2100x2100 (JPEG q=80 / PNG), write to public/base/, and convert the 4 corners with pyproj Transformer.from_crs(25832, 4326, always_xy=True).
 - Write public/base/dop20.json and alkis.json as [{url, bounds: [[lng,lat] BL, TL, TR, BR]]}.
 2. From datasets/osm/altstadt.json, building ways -> public/base/buildings-osm.json as [{polygon: [[lng,lat],...], h}] with h = tags.height, else building:levels*3, else 9. Round coords to 6 dp.
- Done-check: uv run --with pyproj --with pillow python tools/base_assets.py; du -sh public/base (< 15 MB); add tools/check_base.py that asserts 11.07773,49.45095 lies inside exactly one dop20 quadrant and that buildings count >= 3000; run it.

C2 · fixtures (Claude Haiku). {ALLOWED} = src/ui/__fixtures__/.

Goal: realistic fixtures that type-check against src/contracts.ts.

Write src/ui/__fixtures__/gen.ts (run with npx tsx or bun) that reads datasets/osm/altstadt.json and writes:

- candidates.json: 18 Candidate objects. Include the 3 benchmarks (kind 'benchmark': Hauptmarkt 49.45393,11.07744; Lorenzkirche 49.45095,11.07773; Kaufhof ~49.4490,11.0800). At least 5 have passedFilter=false with a rejectReason ("area < 800 m²", "no stop within 400 m", "no van route within 80 m"). Polygons are small quads around real OSM squares.
 - result-<id>-<scenario>.json for 3 shortlist ids x SUNNY_SAT, RAINY_SAT: SimulationResult with all 3 TimeSlices,
 - >= 100 trips whose paths follow OSM footway/pedestrian way geometry with increasing tSec, heat points, bottlenecks.
 - Values are deterministic from a seeded RNG (mulberry32, seed 42). Label the file header "FIXTURE - not engine output".
 - brief.json: one Brief.
- Write fixtures.test.ts that imports every JSON with the contract types (satisfies) and asserts trips.length >= 100.
- Done-check: npm test -- --run src/ui/__fixtures__ ; npx tsc --noEmit.

C4 · map (Claude Opus). {ALLOWED} = src/ui/map/, e2e/map.spec.ts.

Goal: src/ui/map/MapView.tsx (default export) renders MapLibre with an empty background style and deck.gl via @deck.gl/mapbox MapboxOverlay ({interleaved: true}). Initial view 49.4525,11.0775 zoom 16 pitch 50.

Layers (one file per layer factory under src/ui/map/layers/): BitmapLayer per entry in /base/dop20.json using its

4-corner bounds; extruded buildings from /data/buildings.json, falling back to /base/buildings-osm.json on 404;

candidate PolygonLayer coloured by rank from the store (rejected = grey 40% alpha, pickable, tooltip shows name and

rejectReason); bollards ScatterplotLayer from OSM barrier=bollard nodes (precompute once, keep in memory).

Click on a candidate sets selectedId via the store. Show the attribution text from IMPLEMENTATION_PLAN.md §3.

Memoize layers on their data so a weight change does not rebuild static layers.

Done-check: npm run build; npm run e2e -- map.spec.ts (spec: no console errors, canvas present, hover a rejected

candidate shows the reason, saves e2e/__shots__/map.png).

C5 · ranking (Codex). {ALLOWED} = src/ui/ranking/, e2e/ranking.spec.ts.

Goal: src/ui/ranking/RankingPanel.tsx (default export) and applyWeights.ts.
applyWeights(criteria: Weights, w: Weights) returns the weighted sum; normalize(w) renormalizes to sum 1.
Bars sorted by score from store.results[*].criteria and store.weights; 5 sliders (accessibility, footfall, fairness, localBusiness, walkability) and 3 presets exactly as IMPLEMENTATION_PLAN.md §6. Guard badge when any persona score < 40.
Use React keys by candidate id and CSS transform transitions for reordering. No layout thrash: no getBoundingClientRect per frame.
Done-check: npm test -- --run src/ui/ranking (unit tests for normalize, sorting, guard);
npm run e2e -- ranking.spec.ts measuring input->DOM reorder with MutationObserver, median of 20 < 100 ms.

C6 · cockpit (Claude Sonnet). {ALLOWED} = src/ui/cockpit/, e2e/cockpit.spec.ts.

Goal: src/ui/cockpit/Cockpit.tsx (default export): consensus dial (min and mean of the 4 PersonaResult.score of the selected candidate), 4 persona cards (score, topFriction chip, verdict or "verdict pending"), brief panel (why, comparisons, losers), button "Apply Claude mitigation" -> SimClient.run(selectedId, scenario, brief.losers.map(l => l.mitigation), seed) then store update, button "Export council brief" -> Blob download council-brief.md with brief.councilBriefMd.
Done-check: npm run e2e -- cockpit.spec.ts (cards show fixture scores, download text equals councilBriefMd, Apply triggers exactly one adapter call and re-renders).

C7 · sim layers and perf (Codex). {ALLOWED} = src/ui/map/simLayers.ts, src/ui/controls/, src/ui/perf/, e2e/perf.spec.ts.

Goal: export buildSimLayers(result, slice, timeSec) returning TripsLayer (getPath [lng,lat], getTimestamps tSec, persona colours, trailLength 120, currentTime timeSec), HeatmapLayer from bySlice[slice].heat, bottleneck ScatterplotLayer (radius by severity, tooltip cause+time), kiosk IconLayer or ScatterplotLayer when result.mitigations is non-empty. src/ui/controls/TimeControls.tsx: slice slider (3 stops) and play/pause; one requestAnimationFrame loop updates timeSec. Only the TripsLayer changes per frame; keep data arrays stable (useMemo) so deck.gl does not re-upload buffers. src/ui/perf/fps.ts writes a rolling 1 s fps to window.__fps when location.search has perf=1. MapView integration: the orchestrator adds one line; export what it needs and document it.
Done-check: npm run build; npm run e2e -- perf.spec.ts --headed (median __fps over 10 s >= 55 with 100 trails).

C8 · what-if and compare (Claude Sonnet). {ALLOWED} = src/ui/whatif/, src/ui/compare/, e2e/whatif.spec.ts.

Goal: WhatIfBar.tsx (default export): Sunny / Rainy / Christmas buttons set store.scenario and call SimClient.run for every shortlisted candidate (passedFilter true, plus benchmarks), with a spinner and a disabled state while running. Missing fixture for a scenario shows "scenario not available offline". Compare.tsx: two selected candidates side by side, 5 criteria and 4 persona scores.
Done-check: npm run e2e -- whatif.spec.ts.

Cross-review prompt (other vendor, read-only).

```
Review the diff of branch {BRANCH} against main for: fields not present in src/contracts.ts, hard-coded scores,
files edited outside {ALLOWED}, new dependencies, secrets or .env access, deck.gl layers rebuilt every render,
and broken done-check specs. Report findings with file:line. Do not edit files.
```

5.3 Ruddr handoff block

Append the "Work autonomously to completion ... Handoff report" block from `~/claude/skills/ruddr-delegate/SKILL.md` verbatim to every brief.

6. Orchestrator loop

6.1 One-time setup (pre-event)

```
cd "/run/media/maheshk/New Volume/MK-solutions/claude_imapct_lab"
echo ".scratch/" >> .git/info/exclude
ruddr models add claude claude-sonnet-5-5 --label "Claude Sonnet 5.5"
ruddr models --json | grep -c sonnet-5-5          # expect 1
mkdir -p .scratch/c-swarm/{c0,c1,c2,c4,c5,c6,c7,c8}
# brief.md per agent = preamble + task block + handoff block
```

6.2 Launch (example: the 13:30 wave)

```
R="/run/media/maheshk/New Volume/MK-solutions/claude_imapct_lab"
for t in c4-map c5-ranking c6-cockpit; do
  git -C "$R" worktree add "$R/../ut-${t%-*}" -b "c/${t}" c/ui
done
( cd "$R/../ut-c4" && npm ci ) & ( cd "$R/../ut-c5" && npm ci ) & ( cd "$R/../ut-c6" && npm ci ) & wait

ruddr run --detach --provider claude --model claude-opus-5-5 --effort high \
  --cwd "$R/../ut-c4" --sandbox workspace-write --turn-timeout 75m \
  --prompt-file "$R/.scratch/c-swarm/c4/brief.md" --state-dir "$R/.scratch/c-swarm/c4/run" &
ruddr run --detach --provider codex --model gpt-6.1-sol --effort high \
  --cwd "$R/../ut-c5" --sandbox workspace-write --turn-timeout 60m \
  --prompt-file "$R/.scratch/c-swarm/c5/brief.md" --state-dir "$R/.scratch/c-swarm/c5/run" &
ruddr run --detach --provider claude --model claude-sonnet-5-5 \
  --cwd "$R/../ut-c6" --sandbox workspace-write --turn-timeout 60m \
  --prompt-file "$R/.scratch/c-swarm/c6/brief.md" --state-dir "$R/.scratch/c-swarm/c6/run" &
wait
ruddr status --root "$R/.scratch/c-swarm"          # every run must show active
```

Notes: - Always pass `--sandbox workspace-write`. Ruddr defaults to `danger-full-access`, and `~/codex/config.toml` also sets full access with `approval_policy = "never"`. - `workspace-write` under Codex blocks network, so `npm ci` runs from the orchestrator before launch. - Playwright starts a dev server, and parallel worktrees would collide on port 5173. C0 makes `playwright.config.ts` read `process.env.PORT ?? 5173` for both `baseURL` and `webServer` (`vite --port $PORT --strictPort`). Each agent runs its specs with `PORT=517N`, where N is the digit of its ID (C4 → 5174 ... C8 → 5178).

Fallback without Ruddr (not steerable):

```
# Codex
codex exec -C "$R/../ut-c5" -m gpt-6.1-sol -c model_reasoning_effort="high" -s workspace-write \
  --json -o "$R/.scratch/c-swarm/c5/last.md" - < "$R/.scratch/c-swarm/c5/brief.md" > "$R/.scratch/c-
swarm/c5/events.jsonl"
# Claude (cd into the worktree first; -p reads the prompt from stdin)
cd "$R/../ut-c6" && claude -p --model claude-sonnet-5-5 --effort medium --permission-mode acceptEdits \
  --max-budget-usd 4 --output-format json < "$R/.scratch/c-swarm/c6/brief.md" > "$R/.scratch/c-
swarm/c6/out.json"
```

`codex exec` and `claude -w <name>` also create managed worktrees, but they pick their own paths. The plan uses explicit `git worktree add` so the merge commands stay predictable.

6.3 Monitor and steer

```
ruddr tui --root "$R/.scratch/c-swarm" # live view in a side terminal
ruddr peek --root "$R/.scratch/c-swarm" # last trace lines per agent
ruddr wait --root "$R/.scratch/c-swarm" --any --timeout 10m # returns when the next one finishes
ruddr steer --state-dir "$R/.scratch/c-swarm/c5/run" "store.ts now has field X; rebase on c/ui"
ruddr interrupt --state-dir "$R/.scratch/c-swarm/c4/run" # wrong premise
```

Check every agent in its first 2 minutes (peek shows file reads, not an error loop), then every 10 minutes. A silent `trace.log` for more than 10 minutes counts as a stall.

6.4 Verify independently (never trust the handoff report)

For each finished agent:

```
cd "$R/../ut-c5"
git status --short # only files under {ALLOWED}?
git diff --name-only | grep -vE '^(src/ui/ranking/|e2e/ranking.spec.ts)' && echo "SCOPE VIOLATION"
git diff | grep -nE 'sk-ant|ANTHROPIC|process\.env\.[A-Z_]*KEY' && echo "SECRET?"
npx tsc --noEmit && npm test -- --run && PORT=5175 npm run e2e -- ranking.spec.ts
```

Then run the cross-review from the other vendor. `codex review --help` lists `--base <BRANCH>` and a positional `[PROMPT]`; if this version rejects the combination, drop the prompt and keep `--base`.

```
# Claude-written branch reviewed by Codex
git -C "$R/../ut-c6" add -A && git -C "$R/../ut-c6" commit -m "c6: cockpit"
cd "$R/../ut-c6" && codex review --base c/ui "$(cat "$R/.scratch/c-swarm/review.md")"
# Codex-written branch reviewed by Claude, read-only
cd "$R/../ut-c5" && git diff c/ui... | claude -p --model claude-sonnet-5-5 --permission-mode plan \
  --max-budget-usd 1 "$(cat "$R/.scratch/c-swarm/review.md")"
```

Mahesh fixes or steers on any real finding, then merges:

```
git -C "$R" switch c/ui && git -C "$R" merge --no-ff c/c5-ranking
cd "$R" && npm ci && npm run build && npm test -- --run && npm run e2e
git -C "$R" worktree remove "$R/../ut-c5"
```

6.5 Failure policy

Situation	Action
Agent stalls (no trace for 10 min)	<code>ruddr interrupt</code> , then <code>ruddr run --resume-thread <id></code> with a brief that states what it already learned.
Done-check fails after the agent claims success	<code>ruddr steer</code> with the verbatim failing output. This is attempt 2.

Situation	Action
Fails twice	Re-scope (cut the Should part), or switch vendor with the same brief plus the two failure logs, or Mahesh does it by hand if it is under 30 minutes of work. After 15:00, prefer by hand.
Scope violation	Discard the out-of-scope files with <code>git checkout -- <file></code> and steer once. A second violation ends the run.
Agent edits <code>contracts.ts</code>	Revert and steer. Contract changes go only through B and D at the 13:20 freeze.

6.6 Budget

Codex runs use the ChatGPT-plan quota of the `codex` login (the plan tier was not checked here). Claude runs cost API or plan usage. Caps below apply through `--max-budget-usd` when an agent runs via `claude -p`. Ruddr has no budget flag, so Mahesh watches Claude runs in `ruddr tui` and interrupts at the time limit.

Agent	Est. tokens (in+out)	Cap
C0 Codex high	1.5 M	45 min
C1 Sonnet	0.6 M	\$3
C2 Haiku	0.8 M	\$1
C4 Opus high	3 M	\$15, 75 min
C5 Codex high	1.5 M	60 min
C6 Sonnet	1 M	\$4
C7 Codex high	2 M	60 min
C8 Sonnet	0.8 M	\$3
Cross-reviews (×7)	0.3 M each	\$1 each (Claude)
C10 Opus interactive	3 M	\$15
Total Claude		≈ \$50 hard ceiling

7. Integration at 17:00

Step	What	Check
1	B exports a worker entry (<code>src/sim/worker.ts</code>) that takes <code>{candidateId, scenario, mitigations, seed}</code> and posts a <code>SimulationResult</code> . C writes <code>adapters/worker.ts</code> with <code>new Worker(new URL('../sim/worker.ts', import.meta.url), {type: 'module'})</code> .	<code>demo.spec.ts</code> with <code>?data=worker</code> gets 3 results in < 1.5 s each.
2	D exposes <code>/api/brief</code> (POST JSON in, Brief out). Vite <code>server.proxy</code> points <code>/api</code> to D's server port. C writes <code>adapters/http.ts</code> . On error or > 6 s it falls back to D's cached brief, then to the fixture brief, and shows a "cached" chip.	Unplug the network: the brief still renders with the chip.
3	Replace <code>applyWeights</code> with D's score export, keeping the same signature through a one-line re-export.	<code>ranking.spec.ts</code> still < 100 ms.
4	A's <code>public/data/candidates.json</code> replaces the fixture candidates.	Rejected candidates appear grey with A's reasons.
5	Full pitch path (§11) 3 times in a row.	18:00 freeze gate.

Fallbacks. If B is late, the UI stays on `?data=fixtures` for simulation and uses A's real candidates; trails come from fixtures labeled "fixture". If D is late, the UI shows the fixture brief with the "cached" chip and computes `MarketScore` with `applyWeights`. In both cases the URL flag switches back without a rebuild.

8. Risks specific to multi-agent UI work

Risk	Mitigation
Agents invent contract fields	<code>contracts.ts</code> is read-only; <code>npx tsc --noEmit</code> is in every done-check; fixtures use <code>satisfies</code> ; cross-review checks for unknown fields; <code>T0D0(subagent)</code> is the only allowed escape.
Conflicting Tailwind or theme choices	Theme tokens and persona colours live in <code>src/index.css</code> from C0 and are read-only; the preamble lists the hex values; no new CSS files or UI libraries.
deck.gl version mismatch	All <code>@deck.gl/*</code> pinned to one exact version in C0; agents cannot edit <code>package.json</code> ; <code>npm ls @deck.gl/core</code> in the merge check.
Flaky perf measurements	FPS is measured in-app over 10 s, headed, on the demo laptop, median of 3 runs; headless Chromium numbers are ignored. Ranking latency uses the median of 20 changes.
Agents commit secrets	Agents never commit; the preamble forbids <code>.env</code> ; <code>.env</code> is in <code>.gitignore</code> from C0; the merge check greps the diff for keys; Codex runs in <code>workspace-write</code> instead of the config default <code>danger-full-access</code> .
Port and npm collisions between worktrees	One worktree per agent, <code>npm ci</code> per worktree, <code>PORT=517N</code> per spec run.
Merge conflicts in shared files	Shared files are hand-written by Mahesh and read-only for agents; <code>App.tsx</code> mounts stubs from the start.
Misaligned aerial photo	4-corner bounds from <code>pyproj</code> ; C1 check asserts Lorenzkirche falls inside the right quadrant; Mahesh inspects <code>map.png</code> .
Agent spends the window on Stretch work	C9 starts only if C4–C8 are merged by 16:30.